02

MGU BCA - 3RD SEM

Quantitative Techniques

Module 2 - Measures of Central Tendency and Dispersion

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Complete question bank: every important, repeated and sure-mark question for this module, each with a full-mark model answer. Nothing important left out.

EXAM PATTERN (from real MGU question papers)

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

TOPIC 1

Arithmetic Mean

1

The arithmetic mean (average) is the sum of all values divided by the number of values. xbar = (sum of x) / N.

Individual series: add all items and divide by N. Discrete series: xbar = (sum of f*x) / (sum of f).

Continuous series: use the mid-point m of each class; xbar = (sum of f*m) / (sum of f).

The mean uses every value, so very large or very small values pull it up or down.

QUESTION BANK

1 **Define arithmetic mean. How is it calculated for individual, discrete and continuous series?**

IMPORTANT

The arithmetic mean is the most common average. It is the sum of all the values divided by the number of values.

Formula: xbar = (sum of x) / N, where N is the total number of observations.

Individual series: simply add all the items and divide by N.

Example: mean of 4, 6, 8 = (4+6+8)/3 = 18/3 = 6.

Discrete series: when each value x has a frequency f, use xbar = (sum of f*x) / (sum of f).

Multiply each value by its frequency, add these products, and divide by the total frequency.

Continuous series: find the mid-point m of each class interval first.

Then use xbar = (sum of f*m) / (sum of f), where f is the class frequency.

The mid-point is calculated as (lower limit + upper limit) / 2.

Every value in the data is used in this average, so it represents the whole group.

In short, the formula changes only in what we multiply: single values, values times frequency, or mid-points times frequency.

2 **Find the mean of 10, 12, 15, 18, 20.**

IMPORTANT

The formula for an individual series is xbar = (sum of x) / N.

Step 1: add all the values.

sum of x = 10 + 12 + 15 + 18 + 20.

= 22 + 15 + 18 + 20.

= 37 + 18 + 20.

= 55 + 20.

= 75.

Step 2: count the values. N = 5.

Step 3: divide the sum by N.

xbar = 75 / 5.

= 15.

Answer: mean = 15.

3 **What are the merits and demerits of the arithmetic mean?**

IMPORTANT

The arithmetic mean is easy to calculate and simple to understand, even for a beginner.

It uses every value in the data, so no information is wasted.

It is rigidly defined, so two people always get the same answer from the same data.

It is useful for further mathematical work like combined means and standard deviation.

Merit summary: simple, stable, and mathematically the most workable average.

The main demerit is that extreme values pull it up or down and spoil it.

Example: mean of 2, 3, 4, 5, 86 is 20, which describes nobody in the group.

It cannot be found if the data is incomplete or has an open-ended class.

It cannot be used for qualitative data like favourite colour or honesty.

So the mean is best when the data is numerical and free from extreme values.

4 **From the following data compute arithmetic mean by direct method: Marks 0-10: 5, 10-20: 10, 20-30: 25, 30-40: 30, 40-50: 20, 50-60: 10 students. (2019)**

IMPORTANT

For a continuous series the direct method uses class mid-points: Mean = sum(fx) / N.

Mid-points x are 5, 15, 25, 35, 45, 55 for the six classes.

Multiply each mid-point by its frequency: 5x5=25, 15x10=150, 25x25=625, 35x30=1050, 45x20=900, 55x10=550.

Total frequency N = 5+10+25+30+20+10 = 100.

Sum of fx = 25+150+625+1050+900+550 = 3300.

Mean = 3300 / 100 = 33.

Answer: the arithmetic mean is 33 marks.

5 **The mean of 20 values is 45. If one of these values is to be taken 64 instead of 46, find the corrected mean. (2024)**

IMPORTANT

This is a corrected-mean problem: one wrong value must be replaced by the correct one.

Sum of the 20 values as first computed = 20 x 45 = 900.

Remove the wrong value 46 and add the correct value 64: corrected sum = 900 - 46 + 64 = 918.

The correct value 64 is larger than the wrong value 46, so the mean rises slightly after correction.

Corrected mean = 918 / 20 = 45.9.

Answer: the corrected mean is 45.9.

TOPIC 2

Median

2

The median is the middle value when data is arranged in order. Half the values lie below it, half above.

Odd N: median = value at position (N+1)/2. Even N: average of values at N/2 and (N/2)+1.

The median is not affected by extreme values, so it suits skewed data like incomes.

QUESTION BANK

6 **Compute median from the following data: Mid-values 115: 6, 125: 25, 135: 48, 145: 72, 155: 116, 165: 60, 175: 38, 185: 22, 195: 3. (2024, 2019)**

REPEATED

When mid-values are given, first reconstruct the classes; the gap between mid-values is 10, so classes are 110-120, 120-130, up to 190-200.

Total frequency N = 6+25+48+72+116+60+38+22+3 = 390, so N/2 = 195.

Cumulative frequencies: 6, 31, 79, 151, 267, 327, 365, 387, 390.

The median class is the first class whose cumulative frequency reaches 195, which is 140-150.

Here l = 140, previous cf = 151, f = 116, class width h = 10.

Median = l + ((N/2 - cf)/f) x h = 140 + ((195-151)/116) x 10 = 140 + 3.79 = 143.79.

Answer: the median is 143.79.

7 **Define median. Find the median of 5, 8, 12, 15, 22.**

IMPORTANT

The median is the middle value of a dataset when all values are arranged in ascending order.

Half the observations lie below the median and half lie above it.

Odd N: median = the value at position (N+1)/2.

Even N: median = average of the values at positions N/2 and (N/2)+1.

Now find the median of 5, 8, 12, 15, 22.

The data is already arranged in order.

N = 5, which is odd.

Position of median = (5+1)/2 = 6/2 = 3rd value.

The 3rd value in 5, 8, 12, 15, 22 is 12.

Answer: median = 12.

8 **Find the median of 4, 8, 10, 14.**

IMPORTANT

The data 4, 8, 10, 14 is already in ascending order.

N = 4, which is even, so the median is the average of the two middle values.

The two middle positions are N/2 = 2nd value and (N/2)+1 = 3rd value.

2nd value = 8 and 3rd value = 10.

Median = (8 + 10) / 2.

= 18 / 2.

= 9.

Answer: median = 9.

9 **Why is the median preferred over the mean for income data?**

IMPORTANT

Income data is highly skewed because a few people earn very high incomes.

These few extreme incomes pull the arithmetic mean upward, far above what most people earn.

The median is not affected by extreme values because it depends only on the middle position.

So the median shows the income of the typical person much better than the mean.

Example: incomes 10, 12, 15, 18, 200 (in thousands) give mean 51 but median 15.

The median 15 clearly describes the common man, while the mean 51 misleads.

That is why the median is the preferred average for income and wealth data.

TOPIC 3

Mode

3

The mode is the value that occurs most often in the data.

A dataset can have one mode, two modes (bimodal) or no mode at all.

The mode is the only average that works for qualitative data like favourite colour.

QUESTION BANK

10 **Define mode. Find the mode of 2, 3, 3, 5, 7, 3.**

IMPORTANT

The mode is the value that occurs most frequently in a dataset.

It is the value around which the maximum number of items cluster.

A dataset may have one mode, more than one mode, or no mode at all.

Now find the mode of 2, 3, 3, 5, 7, 3.

Count how many times each value appears.

2 appears once, 3 appears three times, 5 appears once, 7 appears once.

The value with the highest frequency is 3.

Answer: mode = 3.

11 **When is the mode the best average to use?**

IMPORTANT

The mode is the best average when we want the most common or most popular value.

It is the only average that can be used for qualitative data.

Example: the most popular shirt size, the most liked colour, or the most common shoe size cannot be averaged by mean or median.

It is also useful when we need a quick, rough average without calculation.

Example: a shopkeeper wants to know which size sells most, so he uses the mode.

So use the mode whenever the question is about the most frequent item.

12 **Calculate mode from the following: Marks (Above) 0: 122, 10: 118, 20: 112, 30: 92, 40: 60, 50: 27, 60: 10, 70: 2, 80: 0 students. (2024)**

IMPORTANT

This is a 'marks above' cumulative table; first convert it to ordinary frequencies by subtraction.

0-10: 122-118 = 4; 10-20: 118-112 = 6; 20-30: 112-92 = 20; 30-40: 92-60 = 32.

40-50: 60-27 = 33; 50-60: 27-10 = 17; 60-70: 10-2 = 8; 70-80: 2-0 = 2.

The modal class is 40-50 with the highest frequency 33.

Mode = l + ((f1-f0)/(2f1-f0-f2)) x h = 40 + ((33-32)/(66-32-17)) x 10.

This gives 40 + (1/17) x 10 = 40 + 0.59 = 40.59.

Answer: the mode is 40.59 marks.

TOPIC 4

4

Empirical Relationship between Mean, Median and Mode

For a moderately skewed distribution: Mode = 3*Median - 2*Mean.

This lets you find any one of the three if the other two are known.

It is called empirical because it comes from observation, not from a strict proof.

QUESTION BANK

13 **State the empirical relationship between mean, median and mode.**

IMPORTANT

For a moderately skewed distribution, the three averages are connected by an empirical formula.

Formula: Mode = 3*Median - 2*Mean.

It is called empirical because it comes from observed data, not from a strict mathematical proof.

It works only for moderately skewed distributions.

If any two of the three are known, the third can be found using this relation.

Example: if mean = 20 and median = 18, then mode = 3*18 - 2*20.

= 54 - 40.

= 14.

So the mode is 14.

14 **If mean = 20 and median = 18, find the mode.**

IMPORTANT

The empirical relationship is: Mode = 3*Median - 2*Mean.

Given: mean = 20 and median = 18.

Substitute the values.

Mode = 3*18 - 2*20.

Step 1: multiply. 3*18 = 54.

Step 2: multiply. 2*20 = 40.

Step 3: subtract. 54 - 40 = 14.

Answer: mode = 14.

TOPIC 5

5

Graphical Location of Median and Mode

The median is found graphically from ogives: draw less-than and more-than ogives; they cross at the median.

The mode is found from a histogram: take the tallest bar and join its top corners to the neighbouring bars to locate the modal value.

Graphical methods are useful when exact calculation is long.

QUESTION BANK

15 **How do you locate the median graphically?**

IMPORTANT

Draw the less-than ogive (cumulative frequency plotted against upper class limits).

On the same graph, draw the more-than ogive (cumulative frequency against lower class limits).

Find the point where the two ogives intersect each other.

From that intersection point, drop a vertical line down to the x-axis.

The value on the x-axis where this line touches is the median.

This works because both ogives meet exactly at the N/2th value.

Graphical median is useful when the data is grouped in classes.

16 **How do you locate the mode graphically?**

IMPORTANT

First draw the histogram of the frequency distribution.

Find the tallest bar; it represents the modal class, where the mode lies.

Join the top-left corner of this tallest bar to the top-left corner of the next bar.

Join the top-right corner of the tallest bar to the top-right corner of the previous bar.

The two lines cross at a point above the modal class.

From this crossing point, drop a vertical line down to the x-axis.

The value where the line meets the x-axis is the mode.

This method is used when exact calculation is long.

TOPIC 6

6

Combined Mean and Weighted Mean

Combined mean joins two or more groups: xbar12 = (n1*xbar1 + n2*xbar2) / (n1 + n2).

Weighted mean gives some items more importance: xbar_w = (sum of w*x) / (sum of w).

Use the weighted mean when items are not equally important, like exam marks with different weightage.

QUESTION BANK

17 **Find the weighted mean of prices 10, 20, 30 with weights 2, 3, 5. (2019, 2022)**

REPEATED

The formula for weighted mean is xbar_w = (sum of w*x) / (sum of w).

Given: prices x = 10, 20, 30 with weights w = 2, 3, 5.

Step 1: multiply each price by its weight.

10*2 = 20.

20*3 = 60.

30*5 = 150.

Step 2: add the products. 20 + 60 + 150 = 230.

Step 3: add the weights. 2 + 3 + 5 = 10.

Step 4: divide. 230 / 10 = 23.

Answer: weighted mean = 23.

18 **The following data gives the weekly wages of workers in a firm, their total working hours and the average working hours per worker. Calculate the average weekly wage per worker. Wages Rs 80-100: total hours 168, avg hours 12; 100-120: 170, 10; 120-140: 225, 9; 140-160: 272, 8.5; 160-180: 126, 7; 180-200: 91, 6.5. (2019, 2022)**

REPEATED

This is a weighted-mean problem: first find the number of workers in each wage group.

Workers = total hours / average hours: 168/12 = 14, 170/10 = 17, 225/9 = 25, 272/8.5 = 32, 126/7 = 18, 91/6.5 = 14.

Total workers = 14+17+25+32+18+14 = 120.

Take wage mid-points 90, 110, 130, 150, 170, 190 as the values and worker counts as weights.

Weighted total = 90x14 + 110x17 + 130x25 + 150x32 + 170x18 + 190x14 = 1260+1870+3250+4800+3060+2660 = 16900.

Average weekly wage = 16900 / 120 = 140.83.

Answer: the average weekly wage per worker is Rs. 140.83.

19 **Find the combined mean: group 1 has 30 students with mean 60, group 2 has 20 students with mean 70.**

IMPORTANT

The combined mean merges two groups into one single average, giving each group's mean a weight equal to its size.

Formula: xbar12 = (n1*xbar1 + n2*xbar2) / (n1 + n2), where n1 and n2 are the group sizes and xbar1, xbar2 are their means.

Here n1 = 30, xbar1 = 60 for group 1, and n2 = 20, xbar2 = 70 for group 2.

Substituting the values: xbar12 = (30*60 + 20*70) / (30 + 20).

First multiply: 30*60 = 1800 and 20*70 = 1400.

Then add: 1800 + 1400 = 3200, and total students = 30 + 20 = 50.

So xbar12 = 3200 / 50 = 64.

Answer: combined mean = 64.

The combined mean lies between the two group means, closer to 60 because the larger group of 30 students carries more weight.

20 **Distinguish between combined mean and weighted mean.**

IMPORTANT

The combined mean joins the means of two or more groups into one overall mean.

The weighted mean joins values where each value is given an importance weight.

In the combined mean, the group sizes n1, n2 act as the weights automatically.

In the weighted mean, the weights are chosen by us to show importance.

Formula for combined mean: xbar12 = (n1*xbar1 + n2*xbar2) / (n1 + n2).

Formula for weighted mean: xbar_w = (sum of w*x) / (sum of w).

Example of combined mean: average marks of two classes of different sizes.

Example of weighted mean: exam subjects with different weightage, like 40% for internals and 60% for finals.

So the combined mean merges groups, while the weighted mean stresses important items.

21 **The mean age of a group of 100 children was 9.35 years. The mean age of 25 of them was 8.75 years and that of another 65 was 10.51 years. What was the mean age for the remainder? (2024)**

IMPORTANT

This is a combined-mean problem solved backwards to find the missing group's mean.

Total age of all 100 children = 100 x 9.35 = 935 years.

Total age of the 25 children = 25 x 8.75 = 218.75 years.

Total age of the 65 children = 65 x 10.51 = 683.15 years.

Number in the remainder group = 100 - 25 - 65 = 10 children.

Their total age = 935 - 218.75 - 683.15 = 33.10 years.

Mean age of the remainder = 33.10 / 10 = 3.31 years.

Answer: the mean age of the remaining 10 children is 3.31 years.

TOPIC 7

7

Partition Values: Quartiles, Deciles, Percentiles

Quartiles divide ordered data into 4 equal parts: Q1 (25%), Q2 = median (50%), Q3 (75%).

Deciles divide data into 10 parts (D1 to D9); percentiles into 100 parts (P1 to P99).

In ordered data, Q1 is the (N+1)/4th value and Q3 is the 3*(N+1)/4th value.

QUESTION BANK

22 **Define quartiles, deciles and percentiles.**

IMPORTANT

Quartiles divide the ordered data into 4 equal parts: Q1, Q2, Q3.

Q1 is the first quartile; 25% of values lie below it and 75% above it.

Q2 is the second quartile; it is exactly the median, with 50% on each side.

Q3 is the third quartile; 75% of values lie below it and 25% above it.

Deciles divide the ordered data into 10 equal parts, named D1 to D9.

Example: D5 is the 5th decile, which is also the median.

Percentiles divide the ordered data into 100 equal parts, named P1 to P99.

Example: P50 is the 50th percentile, which is also the median.

So Q2 = D5 = P50 = median, and all these values show how the data is spread.

23 **Find Q1 and Q3 of 3, 7, 8, 12, 15, 18, 21.**

IMPORTANT

The data 3, 7, 8, 12, 15, 18, 21 is already in ascending order.

N = 7.

Formula: Q1 is the value at position (N+1)/4.

Q1 position = (7+1)/4 = 8/4 = 2nd value.

The 2nd value is 7.

Formula: Q3 is the value at position 3*(N+1)/4.

Q3 position = 3*(7+1)/4 = 3*8/4 = 24/4 = 6th value.

The 6th value is 18.

Answer: Q1 = 7 and Q3 = 18.

TOPIC 8

8

Range, Quartile Deviation and Mean Deviation

Range = maximum value - minimum value. It is the simplest measure of dispersion.

Quartile deviation (QD) = (Q3 - Q1) / 2. It covers the middle 50% of data.

Mean deviation (MD) = (sum of |x - average|) / N, taken about the mean or median.

Relative versions (coefficient of range, QD, MD) let you compare datasets.

QUESTION BANK

24 **Find the range and quartile deviation of 3, 7, 8, 12, 15, 18, 21. (2024, 2022, 2019)**

REPEATED

Range = maximum value - minimum value.

Here, maximum = 21 and minimum = 3.

Range = 21 - 3.

= 18.

Answer: range = 18.

Quartile deviation formula: QD = (Q3 - Q1) / 2.

For this data, Q1 = 7 and Q3 = 18 (found from the quartile positions).

QD = (18 - 7) / 2.

= 11 / 2.

= 5.5.

Answer: quartile deviation = 5.5.

25 **Define range, quartile deviation and mean deviation with formulas.**

IMPORTANT

The range is the simplest measure of dispersion.

Formula: Range = maximum value - minimum value.

It shows the total spread of the data in one figure.

The quartile deviation covers only the middle 50% of the data.

Formula: QD = (Q3 - Q1) / 2.

It is half of the inter-quartile range and ignores extreme values.

The mean deviation is the average of the absolute deviations from an average.

Formula: MD = (sum of |x - xbar|) / N, taken about the mean.

Absolute bars mean we ignore the sign and only take the size of each deviation.

Together, range gives total spread, QD gives middle spread, and MD gives average spread.

26 **Find the mean deviation about the mean for 4, 6, 8, 10, 12.**

IMPORTANT

Mean deviation about the mean: MD = (sum of |x - xbar|) / N.

Step 1: find the mean.

xbar = (4 + 6 + 8 + 10 + 12) / 5.

= 40 / 5.

= 8.

Step 2: find the absolute deviation of each value from 8.

|4 - 8| = |-4| = 4.

|6 - 8| = |-2| = 2.

|8 - 8| = |0| = 0.

|10 - 8| = |2| = 2.

|12 - 8| = |4| = 4.

Step 3: add the absolute deviations.

4 + 2 + 0 + 2 + 4 = 12.

Step 4: divide by N = 5.

MD = 12 / 5.

= 2.4.

Answer: mean deviation about the mean = 2.4.

27 **Calculate mean deviation and its co-efficient: Marks 0-10: 4, 10-20: 6, 20-30: 10, 30-40: 20, 40-50: 10, 50-60: 6, 60-70: 4 students. (2022)**

IMPORTANT

Mean deviation about the mean = sum(f x |x - mean|) / N; first find the mean using mid-points.

Mid-points are 5, 15, 25, 35, 45, 55, 65 with frequencies 4, 6, 10, 20, 10, 6, 4; N = 60.

fx values: 20, 90, 250, 700, 450, 330, 260; sum = 2100, so mean = 2100/60 = 35.

Deviations |x-35| are 30, 20, 10, 0, 10, 20, 30.

f|x-mean| values: 120, 120, 100, 0, 100, 120, 120; sum = 680.

Mean deviation = 680/60 = 11.33.

Co-efficient of mean deviation = MD / mean = 11.33/35 = 0.32.

Answer: mean deviation = 11.33 and its co-efficient = 0.32.

28 **Calculate mean deviation from mean from the following data: Age 0-10: 6, 10-20: 5, 20-30: 8, 30-40: 15, 40-50: 7, 50-60: 6, 60-70: 3 persons. (2024)**

IMPORTANT

First find the mean with class mid-points 5, 15, 25, 35, 45, 55, 65 and frequencies 6, 5, 8, 15, 7, 6, 3; N = 50.

fx values: 30, 75, 200, 525, 315, 330, 195; sum = 1670, so mean = 1670/50 = 33.4.

Deviations |x-33.4| are 28.4, 18.4, 8.4, 1.6, 11.6, 21.6, 31.6.

f|x-mean| values: 170.4, 92.0, 67.2, 24.0, 81.2, 129.6, 94.8; sum = 659.2.

Mean deviation = 659.2/50 = 13.18.

Answer: the mean deviation from mean is 13.18 years.

TOPIC 9

Standard Deviation, Variance and Coefficient of Variation

9

Standard deviation: SD = sqrt(sum((x - xbar)^2) / N). It is the best and most used measure of dispersion.

Variance = (SD)^2 = sum((x - xbar)^2) / N.

Coefficient of variation: CV = (SD / xbar) * 100. It compares the spread of two datasets; lower CV

means more consistent.

SD is in the same units as the data; variance is in squared units.

QUESTION BANK

29 **An analysis of monthly wages paid to workers in two firms A and B gives: Firm A - 550 workers, average wage 50, SD 90; Firm B - 650 workers, average wage 45, SD 120. (1) Which firm pays larger total monthly wages? (2) Find the combined mean and combined SD. (3) In which firm is there greater variability? (2024, 2019, 2022)**

SURE MARK

This is the most repeated QT numerical: combined mean, combined SD and coefficient of variation together.

(1) Total wages: Firm A = 550 x 50 = 27,500; Firm B = 650 x 45 = 29,250. So Firm B pays the larger total monthly wage.

(2) Combined mean = (27500 + 29250) / 1200 = 56750/1200 = 47.29.

For combined SD use: variance = [n1(SD1^2 + d1^2) + n2(SD2^2 + d2^2)] / (n1+n2), where d1, d2 are deviations of group means from the combined mean.

d1 = 50 - 47.29 = 2.71 and d2 = 45 - 47.29 = -2.29.

Numerator = 550 x (8100 + 7.34) + 650 x (14400 + 5.24) = 4459039 + 9363409 = 13822448.

Combined variance = 13822448 / 1200 = 11518.71, so combined SD = sqrt(11518.71) = 107.33.

(3) CV of A = (90/50) x 100 = 180%; CV of B = (120/45) x 100 = 266.67%. Greater variability is in Firm B.

Answer: (1) Firm B; (2) combined mean 47.29, combined SD 107.33; (3) Firm B is more variable.

30 **What is the coefficient of variation? If SD = 4 and mean = 20, find CV. (2024, 2018)**

REPEATED

The coefficient of variation is a relative measure of dispersion.

Definition: it expresses the standard deviation as a percentage of the mean.

Formula: CV = (SD / xbar) * 100.

It has no units, so it can compare two datasets with different units or means.

A lower CV means the data is more consistent; a higher CV means more variation.

Now find CV when SD = 4 and mean = 20.

CV = (4 / 20) * 100.

= 0.2 * 100.

= 20.

Answer: CV = 20%.

31 Define standard deviation and variance with formulas.

IMPORTANT

The standard deviation measures how far the values are spread from the mean.

Definition: SD is the square root of the average of the squared deviations from the mean.

Formula: SD = sqrt(sum((x - xbar)^2) / N).

It is the best and most widely used measure of dispersion.

SD is measured in the same units as the original data.

The variance is simply the square of the standard deviation.

Formula: Variance = sum((x - xbar)^2) / N = (SD)^2.

Variance is measured in squared units, like square rupees.

So SD gives spread in original units, and variance gives spread in squared units.

32 Find the SD and variance of 2, 4, 6, 8, 10.

IMPORTANT

Step 1: find the mean.

xbar = (2 + 4 + 6 + 8 + 10) / 5.

= 30 / 5.

= 6.

Step 2: find the squared deviation of each value from 6.

(2 - 6)^2 = (-4)^2 = 16.

(4 - 6)^2 = (-2)^2 = 4.

(6 - 6)^2 = 0^2 = 0.

(8 - 6)^2 = 2^2 = 4.

(10 - 6)^2 = 4^2 = 16.

Step 3: add the squared deviations.

16 + 4 + 0 + 4 + 16 = 40.

Step 4: variance = sum of squared deviations / N = 40 / 5 = 8.

Step 5: SD = sqrt(variance) = sqrt(8).

sqrt(8) = 2.83 approx.

Answer: variance = 8 and SD = 2.83.

33 Two factories have CV 12% and 18%. Which is more consistent?

IMPORTANT

The coefficient of variation measures consistency: lower CV means less variation and more consistency.

Factory 1 has CV = 12%, factory 2 has CV = 18%.

12% is smaller than 18%, so factory 1 has less relative variation.

Less variation means the performance of factory 1 is more stable and uniform.

Therefore the factory with CV 12% is more consistent.

Answer: the factory with CV 12% is more consistent.

34 If n = 10, sum x = 60, sum x-square = 1,000, calculate standard deviation. (2024)

IMPORTANT

When only the sums are given, use the formula SD = sqrt[(sum x-square / n) - (sum x / n)^2].

sum x-square / n = 1000/10 = 100.

sum x / n = 60/10 = 6, and its square is 36.

Variance = 100 - 36 = 64.

SD = sqrt(64) = 8.

Answer: the standard deviation is 8.